

SERVLET PROGRAMS

1. COOKIES SESSION TRACKING (EXPT NO:1)
- I. preferences.html

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  <title>User Preferences</title>
5
6
7  </head>
8
9  <body>
10
11  <div class="box">
12  <h2>User Preferences</h2>
13
14  <form action="preferences" method="post">
15
16    <select name="theme">
17      <option>Light</option>
18      <option>Dark</option>
19    </select>
20
21    <select name="language">
22      <option>English</option>
23      <option>Tamil</option>
24      <option>Hindi</option>
25    </select>
26
27    <button type="submit">Save Preferences</button>
28
29  </form>
30
31  </div>
32
33  <script>
34  function success() {
35    alert("Preferences saved successfully!");
36    return false;
37  }
38  </script>
39
40  </body>
41  </html>
```

II. Servlet.java

```
package com.cookie;
import java.io.*;

public class Servlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String theme = request.getParameter("theme");
        String language = request.getParameter("language");

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<html><body style='font-family:Arial;text-align:center;'>");
        out.println("<h2 style='color:green;'>Preferences Saved Successfully!</h2>");
        out.println("<p><b>Theme:</b> " + theme + "</p>");
        out.println("<p><b>Language:</b> " + language + "</p>");
        out.println("<a href='preferences.html'>Go Back</a>");
        out.println("</body></html>");
    }
}
```

III. web.xml

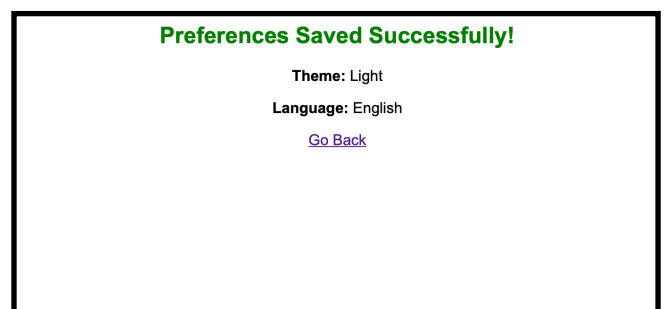
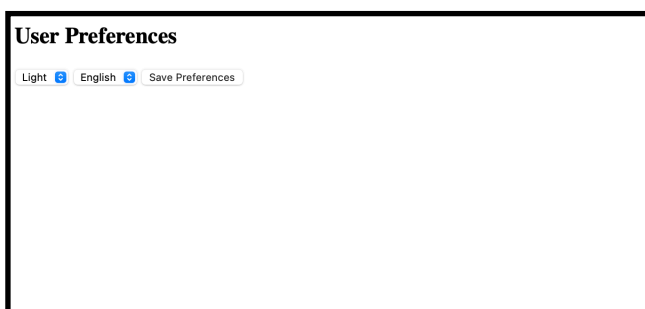
```
Bind to grammar/schema...
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
    version="3.1">

    <servlet>
        <servlet-name>Servlet</servlet-name>
        <servlet-class>com.cookie.Servlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>Servlet</servlet-name>
        <url-pattern>/preferences</url-pattern>
    </servlet-mapping>

</web-app>
```

Output:



2. Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests (EXPT NO:2)

I. login.html

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-scale=1.0">
6     <title>Login Page</title>
7 </head>
8 <body>
9
10     <h2>User Login</h2>
11
12     <form action="LoginServlet.java" method="get">
13
14         <label for="username">Username:</label>
15         <input type="text" id="username" name="username" required>
16
17         <br><br>
18
19         <label for="password">Password:</label>
20         <input type="password" id="password" name="password" required>
21
22         <br><br>
23
24         <input type="submit" value="Login">
25
26     </form>
27
28     <script>
29         function showAlert(event) {
30             event.preventDefault();
31             alert("Login button clicked!");
32         }
33     </script>
34
35 </body>
36 </html>
```

II. LoginServlet.java

```
1 package log;
2 import java.io.IOException;
3
4 public class LoginServlet extends HttpServlet {
5     protected void doPost(HttpServletRequest request, HttpServletResponse response)
6         throws ServletException, IOException {
7         String username = request.getParameter("username");
8         String password = request.getParameter("password");
9         if (username.equals("admin") && password.equals("1234")) {
10             HttpSession session = request.getSession();
11             session.setAttribute("user", username);
12             response.sendRedirect("welcome");
13         } else {
14             response.setContentType("text/html");
15             PrintWriter out = response.getWriter();
16             out.println("<h3>Invalid Login</h3>");
17             out.println("<a href='login.html'>Try Again</a>");
18         }
19     }
20 }
21 }
```


III. LogoutServlet.java

```
1 package log;
2+ import java.io.IOException;
3
4 public class LogoutServlet extends HttpServlet {
5     protected void doGet(HttpServletRequest request, HttpServletResponse response)
6     throws ServletException, IOException {
7         HttpSession session = request.getSession(false);
8         if (session != null) {
9             session.invalidate();
10        }
11        response.sendRedirect("login.html");
12    }
13 }
```

IV. WelcomeServlet.java

```
1 package log;
2+ import java.io.IOException;
3
4 public class WelcomeServlet extends HttpServlet {
5     protected void doGet(HttpServletRequest request, HttpServletResponse response)
6     throws ServletException, IOException {
7         HttpSession session = request.getSession(false);
8         response.setContentType("text/html");
9         PrintWriter out = response.getWriter();
10        if (session != null && session.getAttribute("user") != null) {
11            String user = (String) session.getAttribute("user");
12            out.println("<h2>Welcome, " + user + "</h2>");
13            out.println("<a href='logout'>Logout</a>");
14        } else {
15            out.println("<h3>Session Expired. Please Login Again.</h3>");
16            out.println("<a href='login.html'>Login</a>");
17        }
18    }
19 }
```

V. web.xml

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 Bind to grammar/schema...
3 <web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
4 version="3.1">
5     <servlet>
6         <servlet-name>LoginServlet</servlet-name>
7         <servlet-class>log.LoginServlet</servlet-class>
8     </servlet>
9     <servlet-mapping>
10        <servlet-name>LoginServlet</servlet-name>
11        <url-pattern>/login</url-pattern>
12    </servlet-mapping>
13     <servlet>
14        <servlet-name>WelcomeServlet</servlet-name>
15        <servlet-class>log.WelcomeServlet</servlet-class>
16    </servlet>
17     <servlet-mapping>
18        <servlet-name>WelcomeServlet</servlet-name>
19        <url-pattern>/welcome</url-pattern>
20    </servlet-mapping>
21     <servlet>
22        <servlet-name>LogoutServlet</servlet-name>
23        <servlet-class>log.LogoutServlet</servlet-class>
24    </servlet>
25     <servlet-mapping>
26        <servlet-name>LogoutServlet</servlet-name>
27        <url-pattern>/logout</url-pattern>
28    </servlet-mapping>
29 </web-app>
```

Output:

1. Login Page

User Login

Username:

Password: 

2. Logout Page

Welcome, admin

[Logout](#)

3. JSP (GET & POST Methods)

I. index.jsp

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<!DOCTYPE html>
<html>
<head>
<title>JSP GET and POST Example</title>
</head>
<body>
<h2>JSP Form using GET Method</h2>
<form method="get" action="index.jsp">
Name: <input type="text" name="gname" required><br><br>
Age: <input type="number" name="gage" required><br><br>
<input type="submit" value="Submit using GET">
</form>
<hr>
<h2>JSP Form using POST Method</h2>
<form method="post" action="index.jsp">
Name: <input type="text" name="pname" required><br><br>
Age: <input type="number" name="pageValue" required><br><br>
<input type="submit" value="Submit using POST">
</form>
<hr>
<h2>Submitted Data</h2>
<%
String gname = request.getParameter("gname");
String gage = request.getParameter("gage");
String pname = request.getParameter("pname");
String pageValue = request.getParameter("pageValue");
if (gname != null && gage != null) {
%>
<p><b>GET Method Output:</b></p>
Name: <%= gname %><br>
Age: <%= gage %><br>
<%
}
if (pname != null && pageValue != null) {
%>
<p><b>POST Method Output:</b></p>
Name: <%= pname %><br>
Age: <%= pageValue %><br>
<%
}
%>
</body>
</html>
```

Output:

JSP Form using GET Method

Name:

Age:

JSP Form using POST Method

Name:

Age:

Submitted Data

POST Method Output:

Name: Devyesh
Age: 29

JSP Form using GET Method

Name:

Age:

JSP Form using POST Method

Name:

Age:

Submitted Data

GET Method Output:

Name: Devyesh
Age: 19

4. SERVLET (GET & POST Methods)

I. index.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>GET and POST Servlet Demo</title>
5 </head>
6 <body>
7
8 <h2>Form using GET Method</h2>
9 <form action="hello" method="get">
10 Name: <input type="text" name="name"><br><br>
11 Age: <input type="number" name="age"><br><br>
12 <input type="submit" value="Submit using GET">
13 </form>
14
15 <hr>
16
17 <h2>Form using POST Method</h2>
18 <form action="DoPostServletDemo" method="post">
19 Name: <input type="text" name="name"><br><br>
20 Age: <input type="number" name="age"><br><br>
21 <input type="submit" value="Submit using POST">
22 </form>
23
24 </body>
25 </html>
```

II. DoPostServlet.java

```
1 package com.dopost;
2 import java.io.IOException;
3 import java.io.PrintWriter;
4 import jakarta.servlet.ServletException;
5 import jakarta.servlet.annotation.WebServlet;
6 import jakarta.servlet.http.HttpServlet;
7 import jakarta.servlet.http.HttpServletRequest;
8 import jakarta.servlet.http.HttpServletResponse;
9 public class DoPostServletDemo extends HttpServlet {
10     protected void doGet(HttpServletRequest request, HttpServletResponse response)
11         throws ServletException, IOException {
12         response.setContentType("text/html");
13         PrintWriter out = response.getWriter();
14         String name = request.getParameter("name");
15         String age = request.getParameter("age");
16         out.println("<html><body>");
17         out.println("<h2>GET Method Output</h2>");
18         out.println("Name: " + name + "<br>");
19         out.println("Age: " + age + "<br>");
20         out.println("<a href='index.html'>Back</a>");
21         out.println("</body></html>");
22     }
23     protected void doPost(HttpServletRequest request, HttpServletResponse response)
24         throws ServletException, IOException {
25         response.setContentType("text/html");
26         PrintWriter out = response.getWriter();
27         String name = request.getParameter("name");
28         String age = request.getParameter("age");
29         out.println("<html><body>");
30         out.println("<h2>POST Method Output</h2>");
31         out.println("Name: " + name + "<br>");
32         out.println("Age: " + age + "<br>");
33         out.println("<a href='index.html'>Back</a>");
34         out.println("</body></html>");
35     }
36 }
```


III. web.xml

```
1 <?xml version="1.0" encoding="UTF-8"?>
  Bind to grammar/schema...
2 <web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
3   version="3.1">
4   <servlet>
5     <servlet-name>DoPostServletDemo</servlet-name>
6     <servlet-class>com.dopost.DoPostServletDemo</servlet-class>
7   </servlet>
8   <servlet-mapping>
9     <servlet-name>DoPostServletDemo</servlet-name>
10    <url-pattern>/hello</url-pattern>
11  </servlet-mapping>
12 </web-app>
```

Output:

Form using GET Method

Name:

Age:

Form using POST Method

Name:

Age:

GET Method Output

Name: devyesh
Age: 19
[Back](#)

POST Method Output

Name: devyesh
Age: 19
[Back](#)

5. HELLO WORLD SERVLET

I. HelloServlet.java

```
1 package com.hello;
2 import java.io.IOException;
3 import java.io.PrintWriter;
4 import jakarta.servlet.ServletException;
5 import jakarta.servlet.http.HttpServlet;
6 import jakarta.servlet.http.HttpServletRequest;
7 import jakarta.servlet.http.HttpServletResponse;
8 public class HelloServlet extends HttpServlet {
9     protected void doGet(HttpServletRequest request, HttpServletResponse response)
10     throws ServletException, IOException {
11         response.setContentType("text/html");
12         PrintWriter out = response.getWriter();
13         out.println("<html>");
14         out.println("<body>");
15         out.println("<h1>Hello, World!</h1>");
16         out.println("</body>");
17         out.println("</html>");
18     }
19 }
```

II. web.xml

```
https://jakarta.ee/xml/ns/jakartaee/web-app_5_0.xsd (xsi:schemaLocation with catalog)
1 <?xml version="1.0" encoding="UTF-8"?>
2 <web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
3         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4         xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
5         https://jakarta.ee/xml/ns/jakartaee/web-app_5_0.xsd"
6         version="5.0">
7
8     <servlet>
9         <servlet-name>HelloServlet</servlet-name>
10        <servlet-class>com.hello.HelloServlet</servlet-class>
11    </servlet>
12
13    <servlet-mapping>
14        <servlet-name>HelloServlet</servlet-name>
15        <url-pattern>/hello</url-pattern>
16    </servlet-mapping>
17
18 </web-app>
```

Output:

Hello, World!

6. AJAX (Fetching data from Text File)

I. index.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>AJAX Text File Example</title>
5 <script>
6 function loadStudentData() {
7     var xhr = new XMLHttpRequest();
8     xhr.open("GET", "student.txt", true);
9
10    xhr.onreadystatechange = function () {
11        if (xhr.readyState == 4 && (xhr.status == 200 || xhr.status == 0)) {
12            document.getElementById("output").innerHTML =
13                "<pre>" + xhr.responseText + "</pre>";
14        }
15    };
16
17    xhr.send();
18 }
19 </script>
20 </head>
21 <body>
22 <h2>Student Records (AJAX)</h2>
23 <button onclick="loadStudentData()">Load Student Data</button>
24 <br><br>
25 <div id="output" style="border:1px solid black; padding:10px;"></div>
26 </body>
27 </html>
```

Output:

Student Records (AJAX)

Load Student Data

```
Roll No: 101, Name: Arun, Department: AI&DS
Roll No: 102, Name: Priya, Department: CSE
Roll No: 103, Name: Karthik, Department: IT
Roll No: 104, Name: Divya, Department: ECE
```

7. AJAX WITH JSON

I. index.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>AJAX with JSON</title>
5 </script>
6 function loadStudents() {
7   var xhr = new XMLHttpRequest();
8   xhr.open("GET", "students.json", true);
9
10  xhr.onreadystatechange = function () {
11    if (xhr.readyState == 4 && xhr.status == 200) {
12
13      var students = JSON.parse(xhr.responseText);
14
15      var output = "<table border='1'><tr><th>ID</th><th>Name</th><th>Department</th></tr>";
16
17      for (var i = 0; i < students.length; i++) {
18        output += "<tr><td>" + students[i].id +
19                  "</td><td>" + students[i].name +
20                  "</td><td>" + students[i].dept + "</td></tr>";
21      }
22
23      output += "</table>";
24      document.getElementById("result").innerHTML = output;
25    }
26  };
27
28  xhr.send();
29 }
30 </script>
31 </head>
32 <body>
33   <h2>AJAX with JSON Example</h2>
34   <button onclick="loadStudents()">Load Student Data</button>
35   <br><br>
36   <div id="result"></div>
37 </body>
38 </html>
```

Output:

AJAX with JSON Example

Load Student Data

ID	Name	Department
101	Arun	AI&DS
102	Meena	CSE
103	Karthik	IT

8. AJAX WITH SERVLET

I. index.html

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>AJAX with Servlet</title>
5 <script>
6 function loadStudents() {
7     var xhr = new XMLHttpRequest();
8
9     xhr.open("GET", "/YourProjectName/students", true);
10
11     xhr.onreadystatechange = function () {
12         if (xhr.readyState == 4 && xhr.status == 200) {
13
14             var students = JSON.parse(xhr.responseText);
15
16             var output = "<ul>";
17
18             for (var i = 0; i < students.length; i++) {
19                 output += "<li>" + students[i].name +
20                     " - " + students[i].dept + "</li>";
21             }
22
23             output += "</ul>";
24             document.getElementById("result").innerHTML = output;
25         }
26     };
27
28     xhr.send();
29 }
30 </script>
31 </head>
32 <body>
33 <h2>AJAX with Servlet Example</h2>
34 <button onclick="loadStudents()">Get Student Data</button>
35 <div id="result"></div>
36 </body>
37 </html>
```

II. StudentServlet.java

```
1 import java.io.*;
2
3 @WebServlet("/students")
4 public class StudentServlet extends HttpServlet {
5
6     protected void doGet(HttpServletRequest request, HttpServletResponse response)
7         throws ServletException, IOException {
8
9         response.setContentType("application/json");
10
11         PrintWriter out = response.getWriter();
12
13         String json = "["
14             + "{\"name\":\"John\",\"dept\":\"CSE\"},"
15             + "{\"name\":\"Alice\",\"dept\":\"IT\"},"
16             + "{\"name\":\"Bob\",\"dept\":\"ECE\"}"
17             + "]";
18
19         out.print(json);
20         out.flush();
21     }
22 }
```